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ASSIGNMENT 02

QUESTION 01:

Host A and B are directly connected with a 100 Mbps link. There is one TCP connection between the two hosts, and Host A is sending to Host B an enormous file over this connection. Host A can send its application data into its TCP socket at a rate as high as 120 Mbps but Host B can read out of its TCP receive buffer at a maximum rate of 50 Mbps. **Describe** the effect of TCP flow control.

ANSWER:**Scenario Overview**

- **Link Bandwidth:** 100 Mbps between Host A and Host B
- **Host A's Sending Capacity:** Up to 120 Mbps
- **Host B's Reading Capacity:** Maximum of 50 Mbps from its TCP receive buffer
- **Connection Type:** A single, persistent TCP connection used to transfer a large file

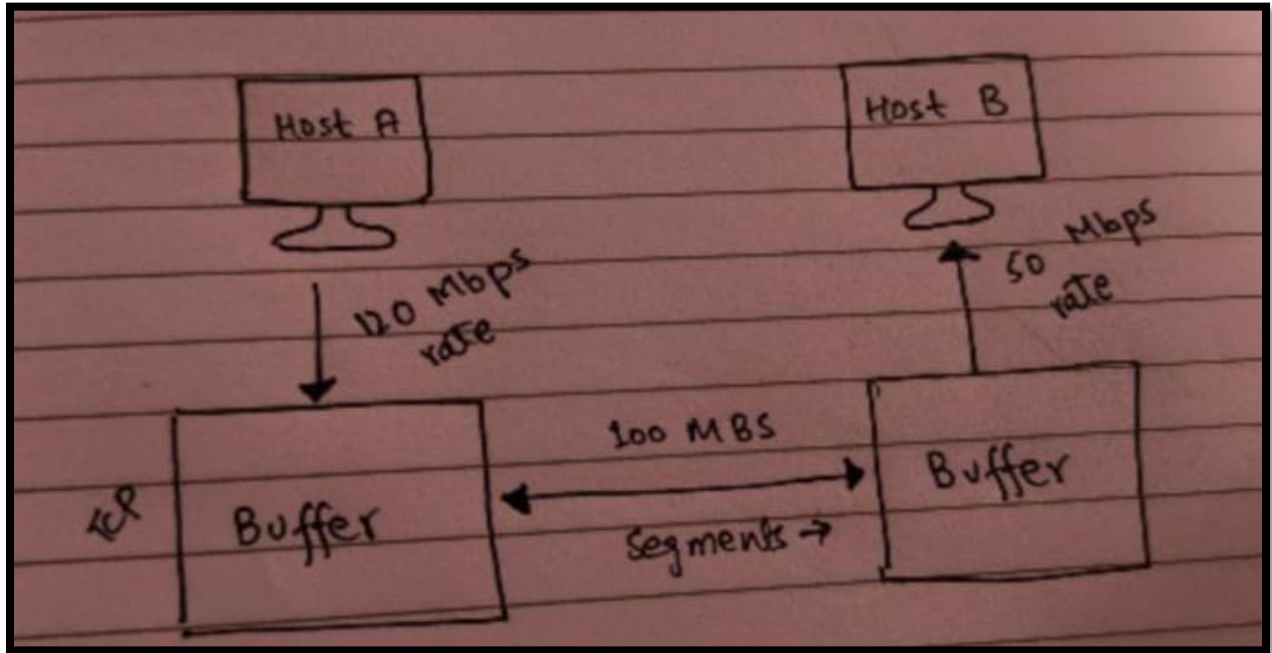
Role of TCP Flow Control

TCP flow control is designed to prevent the sender from overwhelming the receiver. It does this by using the **receiver's advertised window size (rwnd)** to regulate how much data the sender can transmit at any given time.

What Happens in This Setup

- Host A is capable of sending data at 120 Mbps, but the physical link only supports 100 Mbps.
- More critically, Host B can only read data from its buffer at 50 Mbps.
- As Host B processes data slowly, its receive buffer begins to fill.
- Once the buffer nears capacity, Host B reduces its advertised window size—or even sets it to zero—to signal Host A to pause transmission.
- Host A responds by slowing down or temporarily halting data flow until Host B clears space in its buffer.
- This results in a **performance bottleneck**: even though the link could support 100 Mbps, the actual throughput is capped at 50 Mbps due to Host B's limited read rate.

Flow Visualization



Outcome

- TCP flow control dynamically adjusts Host A's transmission rate to match Host B's consumption rate.
- The **effective data rate stabilizes around 50 Mbps**, not the full link capacity or Host A's maximum.
- This mechanism **prevents buffer overflow and data loss**, but also leads to **underutilization of the available bandwidth**.

QUESTION 02:

Compare GBN, SR, and TCP (no delayed ACK). Assume that the timeout values for all three protocols are sufficiently long such that 5 consecutive data segments and their corresponding ACKs can be received (if not lost in the channel) by the receiving host (Host B) and the sending host (Host A) respectively. Suppose Host A sends 5 data segments to Host B, and the 2nd segment (sent from A) is lost. In the end, all 5 data segments have been correctly received by Host B.

- How many segments has Host A sent in total and how many ACKs has Host B sent in total? What are their sequence numbers? Answer this question for all three protocols.
- If the timeout values for all three protocol are much longer than 5 RTT, then which protocol successfully delivers all five data segments in shortest time interval?

ANSWER:**Scenario Overview**

- Host A sends five data segments: sequence numbers 1 through 5
- Segment 2 is lost during transmission
- All five segments are eventually received correctly by Host B
- Timeout duration is long enough to allow all acknowledgments to arrive before any retransmission
- TCP uses immediate acknowledgments (no delayed ACKs)

Protocol Behavior Comparison**1. Go-Back-N (GBN)**

Sender Behavior When the timeout for segment 2 occurs, Host A retransmits segments 2 through 5.

Receiver Behavior Host B only accepts in-order segments. It discards segments 3, 4, and 5 until segment 2 is received.

Transmission Summary

- Segments Sent:
 - Initial: 1, 2, 3, 4, 5
 - Retransmitted: 2, 3, 4, 5
 - Total: 9 segments
- Acknowledgments Sent:
 - ACK 1 (after receiving segment 1)
 - Repeated ACK 1 (for discarded segments 3–5)
 - ACKs 2, 3, 4, 5 (after receiving retransmissions)
 - Total: 7 ACKs
- Sequence Numbers:
 - Segments: 1, 2, 3, 4, 5, 2, 3, 4, 5
 - ACKs: 1, 1, 2, 3, 4, 5

2. Selective Repeat (SR)

Sender Behavior Only the lost segment (segment 2) is retransmitted.

Receiver Behavior Host B accepts and buffers out-of-order segments 3, 4, and 5.

Transmission Summary

- Segments Sent:
 - Initial: 1, 2, 3, 4, 5
 - Retransmitted: 2
 - Total: 6 segments
- Acknowledgments Sent:
 - ACKs for segments 1, 3, 4, 5
 - ACK for segment 2 after retransmission
 - Total: 5 ACKs
- Sequence Numbers:
 - Segments: 1, 2, 3, 4, 5, 2
 - ACKs: 1, 3, 4, 5, 2

3. TCP (No Delayed ACK)

Sender Behavior Retransmits only the missing segment (segment 2), similar to SR.

Receiver Behavior Host B buffers out-of-order segments and sends duplicate ACKs for the last in-order segment.

Transmission Summary

- Segments Sent:
 - Initial: 1, 2, 3, 4, 5
 - Retransmitted: 2
 - Total: 6 segments
- Acknowledgments Sent:
 - ACK 1 (after segment 1)
 - Duplicate ACK 1 (for segments 3, 4, 5)
 - ACK 2 (after retransmitted segment 2)

- ACK 5 (cumulative ACK after all segments received)
- Total: 6 ACKs
- Sequence Numbers:
 - Segments: 1, 2, 3, 4, 5, 2
 - ACKs: 1, 1, 1, 1, 2, 5

Which Protocol Delivers Fastest?

Assuming the timeout is significantly longer than 5 RTTs:

- Selective Repeat and TCP both outperform Go-Back-N because they do not discard segments 3 through 5.
- Go-Back-N delays delivery by waiting for timeout and retransmitting multiple segments.
- TCP (No Delayed ACK) is the fastest in this scenario:
 - It uses fast retransmit, triggered by three duplicate ACKs.
 - This allows segment 2 to be resent before the timeout expires, reducing overall delivery time.

Visualizing Diagram:

